

Supplemental Information

**Affinity Maturation Enhances Antibody Specificity
but Compromises Conformational Stability**

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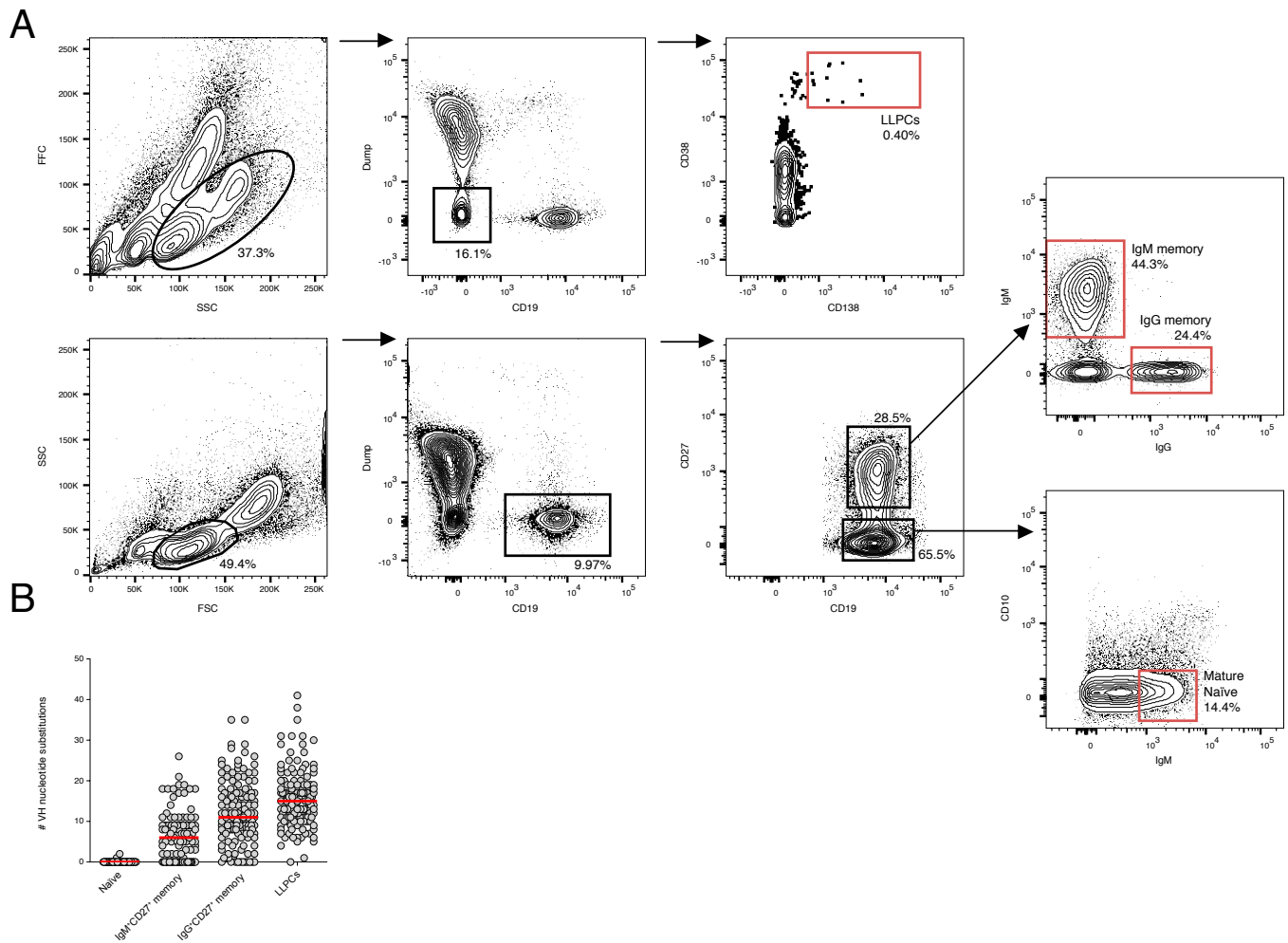


Fig. S1. Representative fluorescence activated cell sorting (FACS) gating strategies for B cell sorting. Related to Figures 1, 2, and 3. (A) Representative sort gates for the isolation of mAbs from LLCs. LLCs were identified by gating on CD19⁺CD38⁺CD138⁺ cells. Bone marrow mononuclear cells from 4 HDs were pooled prior to sorting. (B) Representative sort gates for the isolation of mAbs from naïve B cells, IgM⁺ memory B cells, and IgG⁺ memory B cells. PBMCs were pooled from 3 HDs prior to sorting. Red gates indicate B cell populations that were single-cell sorted. (C) Load of somatic mutations (expressed as number of VH nucleotide substitutions) in mAbs isolated from each B cell subset. Red bars indicate medians.

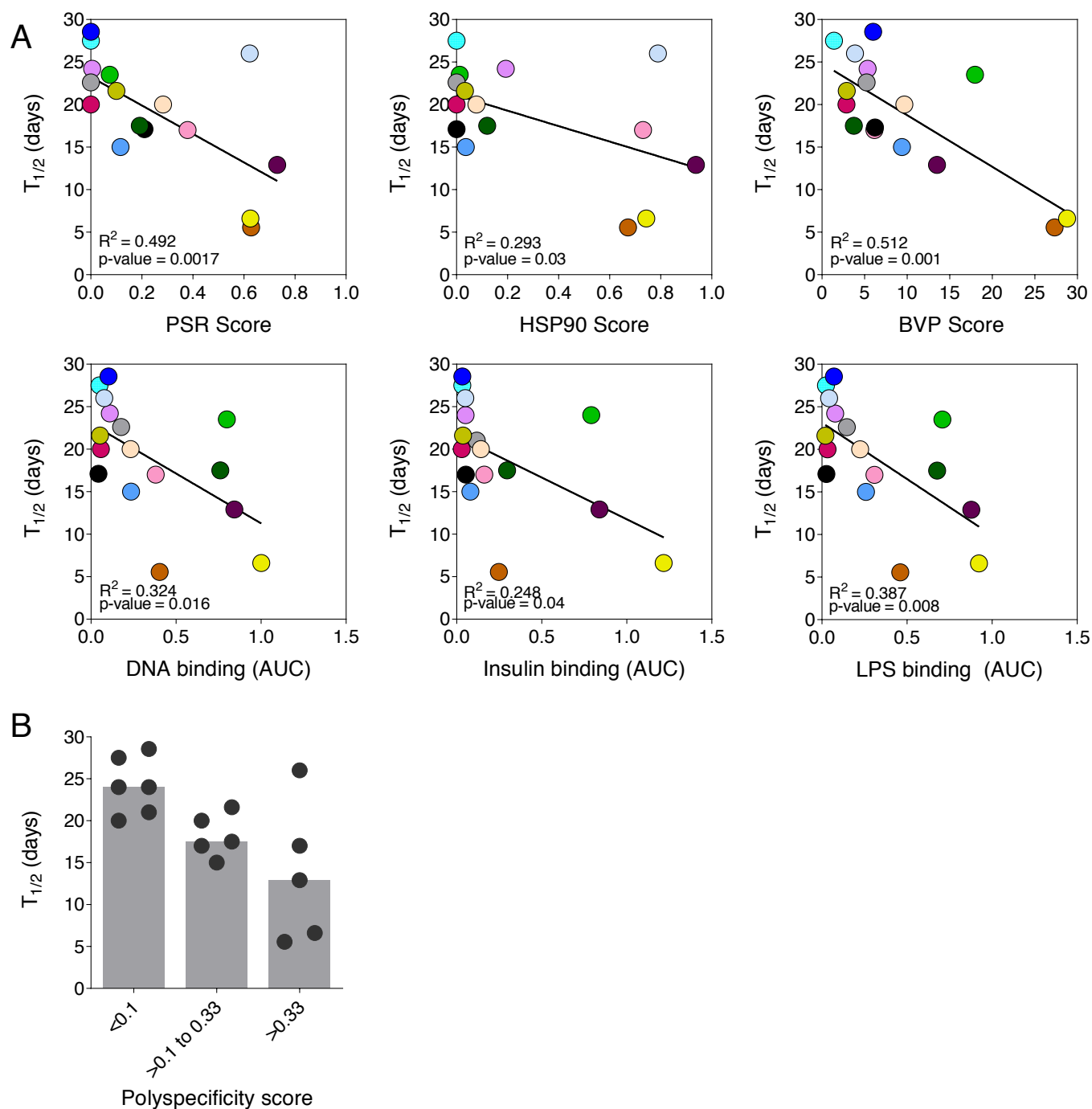


Fig. S2. Assessment of different polyreactivity assay formats. Related to Figures 1 and 5. (A) Correlation between human serum half-life and polyreactivity in different assay formats. Sixteen anti-pathogen mAbs with publicly available phase I clinical trial data were tested for polyreactivity in the indicated assays. Each mAb is shown in a different color. PSR and HSP90 scores were normalized using a panel of control mAbs with high, low, or no detectable polyreactivity. The PSR and HSP90 assays were performed using flow cytometry and the BVP, DNA, insulin, and LPS binding assays were performed by ELISA. R^2 values were determined by linear regression analysis. AUC; area under the curve. (B) PSR assay thresholds used to define high, low, or no polyreactivity. Black dots show the serum half-lives of anti-pathogen mAbs with PSR scores within the indicated ranges. Gray bars show median values. The results shown in (A) and (B) are representative of two independent experiments. See also Table S1.

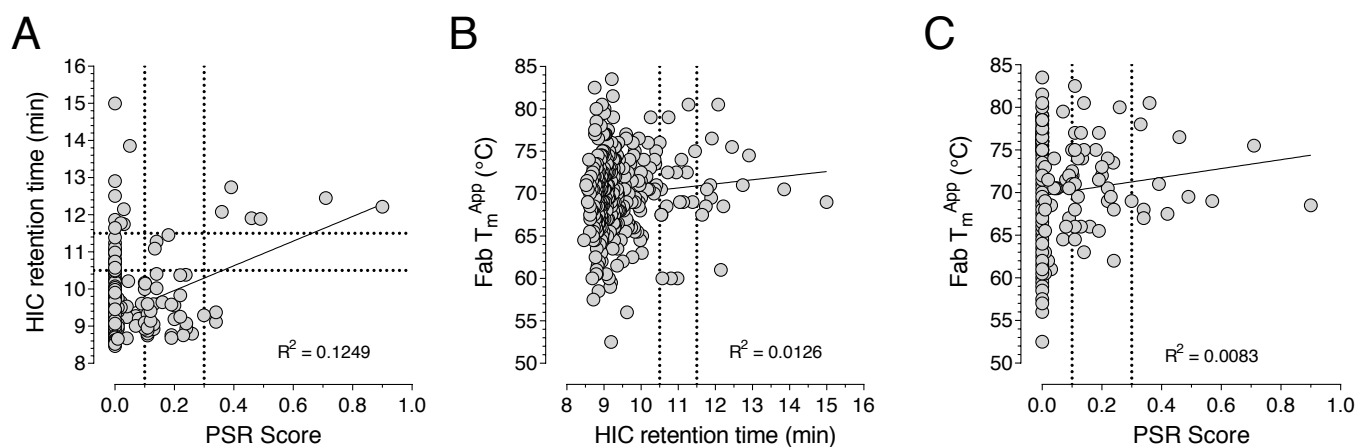


Fig. S3. Related to Figures 1, 2, and 3. Correlation between polyreactivity, hydrophobicity, and thermal stability. (A) Correlation between polyreactivity and hydrophobicity of B cell-derived mAbs. (B) Correlation between thermal stability and hydrophobicity of B cell-derived mAbs. (C) Correlation between thermal stability and polyreactivity of B cell-derived mAbs. Dotted lines show the thresholds for high, medium, and low hydrophobicity and high and low polyreactivity. R^2 values were determined based on linear regression analysis.

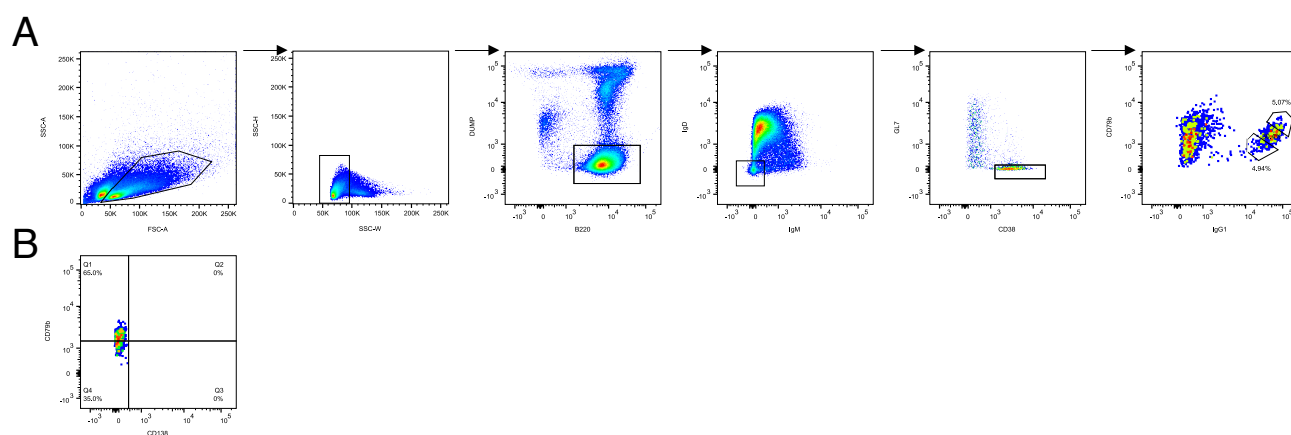


Fig. S4. Related to Figure 4. Representative FACS gating strategy for the isolation of B cells with high and low BCR surface expression. (A) Splenocytes from HEL-immunized mice were pooled and stained with a panel of B cell surface markers to identify IgG₁⁺ B cells with high and low BCR density. The top and bottom 5% of IgG₁⁺CD79b⁺ B cells were single-cell sorted for mAb cloning. (B) B cells from HEL-immunized mice were stained as in (A) but with the addition of an anti-CD138 secondary antibody to calculate the frequency of plasmablasts in the IgG₁⁺CD79b⁺ gate. The plot shown is gated on B220⁺ IgM[−]IgD[−]GL7[−]CD38⁺IgG₁⁺CD79b⁺ B cells.

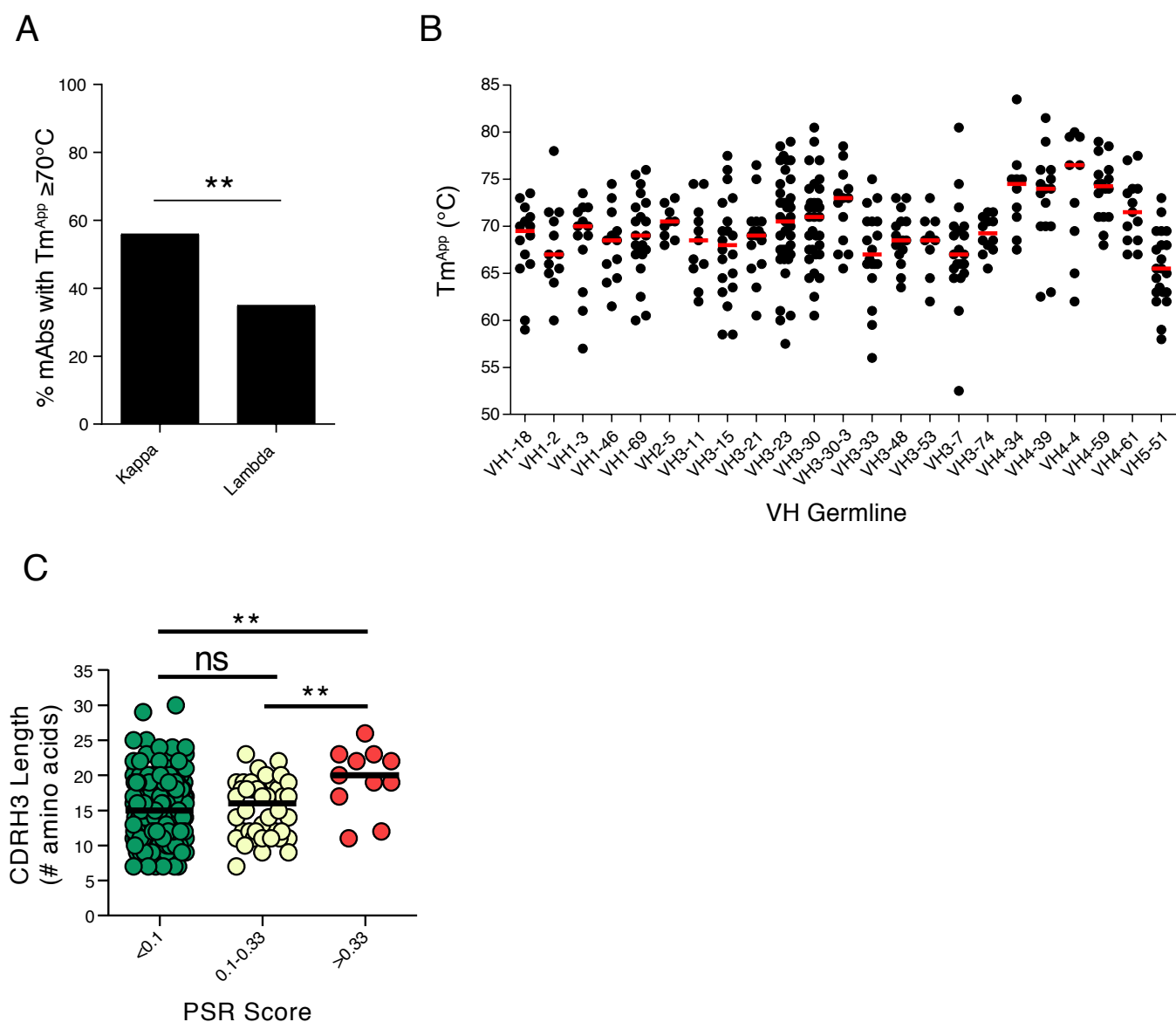


Fig. S5. Related to Figure 5. Effect of LC class, VH germline gene usage, and CDRH3 length on developability properties. (A) Percentage of mAbs with V_κ or V_λ LCs that display $T_m^{App} > 70^\circ\text{C}$. Statistical comparison was made using Fisher's exact test (** $P < 0.01$). (B) T_m^{App} of mAbs grouped by VH germline gene family. Red bars indicate medians. (C) CDRH3 lengths of mAbs with no (PSR score < 0.1), low (PSR score between 0.1-0.33), or high (PSR score > 0.33) polyreactivity. Black bars indicate median values. Statistical comparisons were made using the Mann-Whitney test (** $P < 0.01$).

Table S1. Related to Figure 1. Human serum half-lives and polyreactivity profiles of 16 anti-infectious disease antibodies.

Antibody	Pathogen	Serum half-life (days)	PSR score	Chaperone score	BVP score	dsDNA binding (AUC)	Insulin binding (AUC)	LPS binding (AUC)
2F5	HIV	5.55	0.63	0.67	27.3	0.40	0.25	0.46
4E10	HIV	6.6	0.63	0.74	28.8	1.00	1.22	0.92
10-1074	HIV	24	0.07	0.01	18.0	0.80	0.79	0.71
VRC01	HIV	15	0.12	0.04	9.4	0.24	0.08	0.26
motavizumab	RSV	24	0.01	0.19	5.4	0.11	0.05	0.08
palivizumab	RSV	20	0.00	0.00	2.9	0.06	0.03	0.03
3BNC117	HIV	17	0.38	0.73	6.2	0.38	0.16	0.31
MEDI8852	Influenza	21	0.00	0.00	5.2	0.18	0.12	0.15
VIS410	Influenza	12.9	0.73	0.94	13.5	0.84	0.84	0.88
atidortoxumab	Staph aureaus	21.6	0.10	N/A	2.9	0.05	0.04	0.02
berlimatoxumab	Staph aureaus	27.5	0.00	N/A	1.4	0.05	0.03	0.02
raxibacumab	Anthrax	17	0.21	0.00	6.2	0.04	0.05	0.03
obiltoxaximab	Anthrax	20	0.28	0.08	9.7	0.23	0.14	0.22
urttoxazumab	Shiga-like toxin 2	26	0.62	0.79	3.9	0.08	0.05	0.04
tefibazumab	Staph aureaus	17.5	0.19	0.12	3.7	0.76	0.30	0.68
actoxumab	C. difficile toxin A	28.55	0.00	0.00	6.0	0.10	0.03	0.07